INTERPRETABLE INVERSE ACOUSTIC MODELING OF PERCUSSION VIA PHYSICS-GUIDED SPECTRAL DECOMPOSITION

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ABSTRACT

We propose a physics-based parametric framework for percussive instruments that acts as a *digital twin*, enabling sound synthesis through direct control of physical parameters such as membrane tension, strike position, and geometry. The model generates sound from first principles using eigenmodes of a non-uniform membrane, yielding a physically interpretable representation. We introduce Physics-Guided Spectral Decomposition (PGSD), an inverse method that aligns the model with observed audio via closed-form ridge regression, enabling estimation of modal amplitudes and underlying physical parameters. Experiments on synthetic data demonstrate physically consistent behavior under parameter variation, while evaluation on real Chenda recordings shows accurate reconstruction (coefficient of determination, R^2 up to 0.8152) and consistent excitation trends. The proposed framework provides a unified, bidirectional approach for controllable synthesis and interpretable parameter inference without requiring training data.

1. INTRODUCTION

Controllable sound generation remains a fundamental challenge in music information retrieval (MIR) [1]. While prior work has focused on tasks such as onset detection, classification, and rhythm analysis [2–4], these approaches do not support sound generation under explicit physical constraints or allow direct manipulation of interpretable parameters such as membrane tension, strike position, and geometry. Probabilistic and generative methods have been explored for rhythm and structure [5], but remain largely disconnected from underlying physical processes.

This limitation is particularly evident for percussion instruments, where sound production is governed by complex membrane dynamics arising from the interaction of geometry, material properties, and excitation. The Chenda [6], a South Indian percussion instrument, exhibits rich modal behavior due to non-uniform membrane loading and structural variations (Fig. 1), yet remains largely unexplored in MIR. Early studies established the physics of membrane vibration [7], but have not led to computational frameworks for controllable synthesis and analysis.

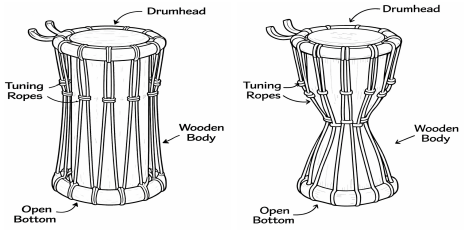


Figure 1: Drum shell geometries: Chenda with cylindrical shell (left), Waisted drum with hourglass shell (right).

In this work, we introduce a *physics-based parametric sound synthesis framework* that functions as a digital twin for membrane-based percussion instruments. We model percussion sound generation using a physics-based formulation in which membrane eigenmodes define a mapping from physical parameters to sound. Crucially, the framework is bidirectional: alongside forward synthesis, we incorporate Physics-Guided Spectral Decomposition (PGSD) to align the model with observed audio, enabling parameter inference and calibration.

Recent work on drum synthesis spans physics-based and data-driven approaches. Modal and transfer-function methods reconstruct acoustics from measured responses [8], while semi-analytical models incorporate structural dynamics [9]. Neural approaches, including DDSP and transformer-based models [10, 11], achieve high-quality generation but rely on training data and offer limited physical interpretability. In contrast, our approach provides a unified, physics-grounded framework for both synthesis and analysis. Unlike prior modal or sinusoidal methods, the proposed framework treats the physical model itself as both the synthesis engine and the analysis basis, yielding a closed-loop, interpretable digital twin. While recent work on physics-informed learning [12] integrates physical constraints into neural models, such methods still rely on training data. Our work advocates a shift from purely statistical MIR approaches toward physically grounded modeling, enabling controllable synthesis, inverse parameter estimation,

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and interpretable analysis within a unified framework.

Contributions: (i) A bidirectional digital twin framework for percussion, unifying forward synthesis and inverse parameter estimation within a shared physics-based representation; (ii) Physics-Guided Spectral Decomposition (PGSD), a training-free inverse method — with closed-form modal-amplitude estimation at its core — that directly recovers physically interpretable parameters from audio; (iii) A demonstration that interpretable physical inference can be achieved without data-driven training, contrasting with prevailing approaches.

2. PHYSICS-BASED SOUND SYNTHESIS MODEL OF THE CHENDA

We model the Chenda as a circular membrane with non-uniform mass loading, enabling a physically grounded description of its vibrational behavior. The overall framework follows a forward modeling approach: given physical parameters (membrane tension, density distribution, and strike position), we compute modal frequencies, damping characteristics, and resulting sound. The membrane is discretized in polar coordinates using a Fourier–Chebyshev collocation scheme [13], which yields a generalized eigenvalue problem incorporating spatially varying density. Solving this system provides modal frequencies and shapes, which form the basis for sound synthesis. First eigenmodes generated from cylindrical and hourglass shell-based drums are given in Fig. 2.

2.1 Membrane Eigenvalue Problem

The Chenda membrane is modeled as a circular membrane of radius $R = 0.125\,\mathrm{m}$, surface tension T, and spatially varying surface density $\rho(\tilde{r}, \theta)$. In normalized polar coordinates ($\tilde{r} = r/R$), the free-vibration problem is given by [7]:

$$\mathbf{L}\,\mathbf{\Psi} = -\lambda^2\,\mathbf{B}\,\mathbf{\Psi}, \qquad \mathbf{B} = \mathrm{diag}(\rho(\tilde{r}, \theta)) \tag{1}$$

where $\mathbf{L}$ is the polar Laplacian discretized using a Fourier–Chebyshev grid (number of radial grid points, $N_r = 25$, number of angular grid points, $N_\theta = 24$). This formulation captures the effect of non-uniform density through the mass matrix $\mathbf{B}$. $\mathbf{\Psi}_n$ denotes the discretized spatial eigenmode (mode shape) of the membrane corresponding to eigenvalue λ_n, defined over the $(\tilde{r}, \theta)$ grid. The eigenvalues λ_n determine the modal frequencies:

$$f_n = \frac{\lambda_n}{2\pi R}\sqrt{\frac{T}{\sigma_{\mathrm{ref}}}} \tag{2}$$

where σ_{ref} is a reference surface density. This establishes a direct mapping from physical parameters to modal frequencies.

2.2 Shell Geometry and Density Models

Different Chenda geometries are modeled through their density distributions $\rho(\tilde{r}, \theta)$. We consider cylindrical, hourglass, and oval configurations, each producing distinct modal structures. In particular, the oval configuration introduces eccentric loading, which breaks circular symmetry and lifts modal degeneracy. This results in richer and more complex spectral characteristics, consistent with observed tonal variations in practice. The shell geometry and simulated fundamental frequencies for a tension of 3500 N/m are shown in Table 1.

2.3 Damping Model

To model energy dissipation, we adopt a frequency-dependent Rayleigh damping formulation [14]. The frequency (ω)-dependent modal decay rate, $\alpha(\omega)$ governing the exponential attenuation of each vibrational mode is defined as,

$$\alpha(\omega) = \frac{\eta_1}{2} + \frac{\eta_2\,\omega^2}{2} \tag{3}$$

where η_1 and η_2 control viscous and structural losses, respectively. Values were selected to match the decay characteristics observed in real Chenda recordings, balancing energy dissipation from viscous damping (η_1) against frequency-dependent structural losses (η_2). This produces mode-dependent decay rates that link physical damping mechanisms to the temporal envelope characteristics of the sound. This formulation yields mode-dependent damping coefficients $\alpha_n = \alpha(\omega_n)$.

2.4 Signal Model

The final sound is synthesized as a sum of damped modal components [15]. For a strike at position (r_s, θ_s):

$$x(t) = \sum_{n=1}^{N} A_n\,e^{-\alpha_n t}\sin(2\pi f_n t) + \varepsilon(t) \tag{4}$$

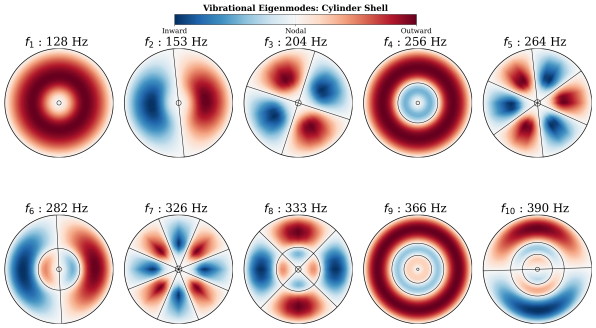
where the modal amplitudes

$$A_n = |\Psi_n(r_s/R,\,\theta_s)| \cdot w_n, \quad w_n = (n+1)^{-0.8} \tag{5}$$

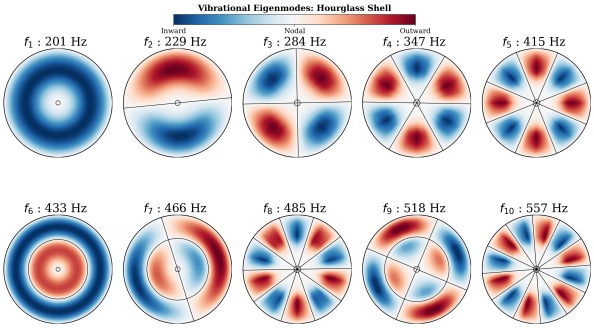
are determined by the mode shapes evaluated at the strike location, with a frequency-dependent radiation weighting w_n. The exponent -0.8 was selected empirically to match the high-frequency roll-off observed in real strike spectra, consistent with the physical expectation that higher-order modes couple less efficiently to the excitation and radiate less efficiently to the far field. $\varepsilon(t)$ represents a residual term accounting for unmodeled dynamics and noise.

This formulation completes the forward model: physical parameters (T, ρ, r_s, θ_s) determine modal frequencies, damping, and amplitudes, which together generate the acoustic signal. The model thus provides a controllable mapping from physical configuration to sound, forming the basis for a digital twin representation of the Chenda [16].

The eigen-problem (Eq. 1) yields a complete orthogonal set of eigenpairs $\{(\lambda_n, \Psi_n)\}_{n=1}^{N}$. Each mode $\Psi_n(r, \theta)$ oscillates at f_n (Eq. 2) and decays at rate $\alpha_n = \alpha(\omega_n)$ (Eq. 3), where $\omega_n = 2\pi f_n$. When the membrane is struck at (r_s, θ_s), each mode is excited with amplitude A_n (Eq. 5).



(a) First 10 eigenmodes of the Cylinder shell



(b) First 10 eigenmodes of the Hourglass shell

Figure 2: First 10 eigenmodes at $T = 3500\,\text{N/m}$ for (a) the Cylinder shell and (b) the Hourglass shell. Color encodes displacement amplitude; nodal patterns and symmetry structure are visible across modes.

Table 1: Shell geometry parameters and fundamental frequencies at $T = 3500\,\text{N/m}$.

Shape	Density profile	f_1 (Hz)
Cylinder	Concentric Gaussian ($\varepsilon = 0$)	127.8
Hourglass	Ring load at $\tilde{r} = 0.55$	201.2
Oval	Eccentric Gaussian ($\varepsilon = 0.18$)	119.3

3. PHYSICS-GUIDED SPECTRAL DECOMPOSITION

Building on the signal formulation in Section 2, Physics-Guided Spectral Decomposition (PGSD) constitutes the inverse component of the proposed framework. While the physical model defines a forward mapping from parameters to sound, PGSD enables calibration and analysis by projecting observed signals onto the same modal basis, consistent with classical sinusoidal and modal decomposition approaches [15, 17]. By incorporating physically derived modal structures [16], the decomposition is constrained to acoustically meaningful components. PGSD thus establishes a unified bidirectional framework by enforcing consistency between simulated and observed acoustics within a shared physical representation. Unlike purely data-driven approaches, it does not require training data and enables physically interpretable parameter estimation. While conceptually related to physics-informed modeling paradigms [12], the proposed method remains fully physics-based.

3.1 Basis Construction

The input signal $x(t)$ may correspond either to a synthesized signal generated by the forward model (Section 2) or to a real recording, enabling both controlled validation and real-world analysis. Given a shell geometry and reference tension T_{ref}, the eigenfrequencies $\{f_n\}$ and decay rates $\{\alpha_n\}$ are computed from Eqs. 2 and 3. The resulting time-domain basis matrix $\mathbf{\Phi} \in \mathbb{R}^{K\times N}$ is defined as [18]:

$$\Phi[k, n] = e^{-\alpha_n t_k} \sin(2\pi f_n t_k) \tag{6}$$

where K is the number of samples and $N = 18$ the number of modes retained for synthesis. This truncation is justified by the amplitude weighting $w_n = (n + 1)^{-0.8}$ (Eq. 5), under which modes beyond $N = 18$ contribute negligible energy to the reconstructed signal; the full eigenbasis (30 modes) is retained for the frequency-domain analysis in Section 2. Here t_k denotes the discrete time samples. $\mathbf{x} = [x(t_1), \ldots, x(t_K)]^\top \in \mathbb{R}^K$ denotes the discretized time-domain signal, where t_k are uniformly sampled time instants. The exponential term models energy decay, while the sinusoidal term captures modal oscillations governed by tension and geometry.

3.2 Ridge Decomposition

Modal amplitudes are estimated via scale-adaptive ridge regression, a standard approach for ill-posed inverse problems [19, 20]:

$$\hat{\mathbf{c}} = (\mathbf{\Phi}^\top \mathbf{\Phi} + \lambda \mathbf{I})^{-1} \mathbf{\Phi}^\top \mathbf{x}, \quad \lambda = \lambda_{\text{rel}} \cdot \|\mathbf{\Phi}^\top \mathbf{\Phi}\|_F \tag{7}$$

with $\lambda_{\text{rel}} = 10^{-6}$. Scaling λ by the Frobenius norm of the Gram matrix ensures consistent regularization across parameter settings and mitigates ill-conditioning from closely spaced modes. Each coefficient $\hat{c}_n$ directly corresponds to the contribution of a physical eigenmode.

3.3 Iterative Frequency Refinement

If the instrument deviates from the reference tension T_{ref}, residual spectral energy remains after decomposition. PGSD iteratively refines modal frequencies to account for this mismatch [21]:

$$f_n \leftarrow (1 - \text{lr})\, f_n + \text{lr} \cdot \underset{f \in \{\, f_n \pm 32\,\text{Hz} \,\}}{\arg\max} |\text{FFT}(\mathbf{x} - \mathbf{\Phi}\hat{\mathbf{c}})| \tag{8}$$

with learning rate $\text{lr} = 0.25$, for up to 14 iterations or until $\Delta R^2 < 10^{-6}$. The search window ($\pm 32\,\text{Hz}$) and learning rate were selected empirically to balance convergence speed against the risk of jumping to a neighboring mode's frequency, particularly in symmetric geometries where modes are closely spaced (Section 4.2.1). At each step, modal frequencies are adjusted toward dominant residual peaks, and decay rates are updated via Eq. 3. This process incrementally aligns the model with observed acoustics without requiring additional supervision.

Defining $\mathbf{\Phi}$ as in Eq. 6, the signal model becomes:

$$\mathbf{x} = \mathbf{\Phi}\,\mathbf{c} + \boldsymbol{\varepsilon} \tag{9}$$

where $\mathbf{c} \in \mathbb{R}^N$ are the modal amplitudes.Given $\mathbf{\Phi}$, this is a standard overdetermined linear system ($K \gg N$), and the modal-amplitude coefficients admit a closed-form solution (Eq. 7). The full inference pipeline is not entirely closed-form, however: frequency refinement (Eq. 8) is an iterative search, and strike-position recovery (Eq. 11) uses template matching over candidate positions. Both remain training-free, requiring no labeled data or learned parameters. The computational cost is $\mathcal{O}(N^2K + N^3)$ with $N = 18$ and $K \approx 33{,}000$, yielding wall-clock times under 0.5 s per strike on a standard laptop. For real recordings, modal frequencies are initialized from prominent spectral peaks to compensate for unknown parameters (e.g., tension or boundary conditions), improving convergence and reconstruction accuracy.

3.4 Diagnostic Inference and Physical Parameter Mapping

The refined modal parameters $(\hat{\mathbf{c}}, \{\hat{f}_n\})$ enable recovery of key physical properties through structured inverse mappings grounded in the modal representation. Since a Chenda strike can be modeled as a superposition of decaying eigenmodes, the signal representation is linear in modal amplitudes and encodes physical parameters in distinct and interpretable components.

Membrane Tension. Tension is encoded in the modal frequencies via the proportionality $f_n \propto \sqrt{T}$ for a fixed mode and geometry. Using the refined fundamental frequency, tension is estimated as:

$$\hat{T} = T_{\text{ref}} \left(\hat{f}_1 / f_{1,\text{ref}} \right)^2 \tag{10}$$

This inversion is exact under the assumed model and does not require calibration data.

Strike Position. Strike location is encoded in the modal amplitude pattern $\{|\Psi_n(r_s/R, \theta_s)|\}$. Due to degeneracy in azimuthally symmetric modes, direct coefficient-based inversion is ill-conditioned. Instead, signal-level template matching is performed over the attack transient (first 500 samples):

$$\hat{r}_s = \arg\max_r \frac{\hat{\mathbf{x}}(r)^\top \mathbf{x}}{\|\hat{\mathbf{x}}(r)\|\,\|\mathbf{x}\|} \tag{11}$$

where $\hat{\mathbf{x}}(r)$ denotes a synthetic signal generated at candidate radial position r, and the objective corresponds to normalized cross-correlation. This approach resolves ambiguity without requiring ground-truth position labels.

Shell Shape Identification. Shell shape is encoded in the eigen-spectrum structure $\{f_n\}$, which varies with the density profile $\rho(\tilde{r}, \theta)$. The signal is decomposed using each candidate shape basis, and the best match is selected via:

$$\hat{s} = \arg\max_s R^2(\mathbf{x},\ \mathbf{\Phi}_s \hat{\mathbf{c}}_s) \tag{12}$$

where $R^2(\cdot,\cdot)$ denotes the coefficient of determination. This corresponds to a model selection procedure, where each basis encodes a distinct physical hypothesis.

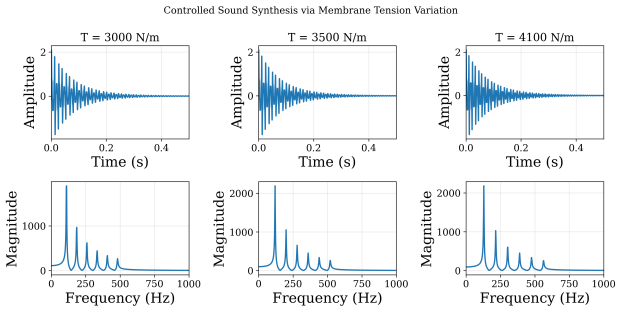


Figure 3: Increasing tension produces systematic upward shifts in modal frequencies ($f \propto \sqrt{T}$), confirming the physical validity and controllability of the model.

4. SIMULATION, VALIDATION, AND RESULTS

4.1 Experimental Setup

All experiments use synthetic signals sampled at $f_s = 44{,}100$ Hz with a duration of 3.0 s and SNR of 30 dB. We evaluate three geometric shapes (Cylinder, Hourglass, Oval) across five membrane tensions ($T \in \{3000, 3200, 3500, 3800, 4100\}$ N/m) and five radial strike positions ($r/R \in \{0, 0.25, 0.50, 0.75, 0.90\}$), resulting in 75 synthetic test signals per task. To assess the controllability of the proposed physics-based synthesis framework, signals are generated by varying membrane tension while keeping geometry and strike position fixed. Fig. 3 illustrates representative waveforms and spectra for different tension values. Increasing tension produces systematic upward shifts in modal frequencies, consistent with the theoretical relationship $f \propto \sqrt{T}$, thereby validating the model's physical consistency and controllability.

4.2 Validation on Synthetic Audio samples

We validate the proposed physics-based synthesis framework and its associated calibration mechanism (PGSD) using synthetic audio signals generated by the physical model described in Section 3. A total of seventy-five samples are generated with varying tension, strike positions, and shapes. Audio samples are generated with known modal frequencies, damping coefficients, and amplitudes, enabling direct comparison between ground-truth and estimated parameters. These parameters are then estimated using the PGSD framework and evaluated quantitatively.

4.2.1 Task 1: Tension Estimation

PGSD achieves high reconstruction accuracy across all synthetic cases, with a mean R^2 exceeding 0.95. Results in Table 2 show that within $\pm 14\%$ of the reference tension T_{ref}, all shapes achieve a mean error below 0.9%. Estimated modal frequencies closely match ground-truth values, with relative errors typically below 2%. The oval geometry is the only shape maintaining high accuracy across the full tension range, with mean error of only 0.32%, because its lifted modal degeneracies eliminate the frequency-search ambiguity that causes mode-jump failures in symmetric geometries. Cylinder and hourglass shapes exhibit mode-jump failures

Table 2: Tension estimation error (%) by shape and tension (N/m). † indicates mode-jump failure.

Shape	Tension (N/m)					Mean
	3000	3200	3500	3800	4100	
Cylinder	0.82	0.78	0.42	9.92†	10.12†	4.41%
Hourglass	0.59	0.42	0.83	6.95†	6.92†	3.14%
Oval	0.76	0.12	0.03	0.54	0.16	**0.32%**

Table 3: Strike position error ($\%R$) by shape and radial position.

Shape	$r = 0$	$0.25R$	$0.50R$	$0.75R$	$0.90R$	Mean
Cylinder	0.0	0.4	0.3	0.1	0.4	**0.3%**R
Hourglass	0.0	0.4	0.3	0.1	0.4	**0.3%**R
Oval	0.0	0.4	0.3	0.1	0.9	**0.3%**R

at higher tensions (3800 N/m and above), resulting in errors exceeding 9%. Mode-jump failures arise from spectral degeneracies in symmetric geometries. In cylindrical and hourglass shapes, multiple eigenmodes share identical or closely spaced frequencies, leading to clustered eigenvalues and ambiguity in the inverse problem. As tension varies, small frequency shifts can reorder these clusters, causing the estimator to switch between nearly equivalent modal solutions and resulting in large errors. In contrast, the oval geometry breaks symmetry and lifts these degeneracies, producing well-separated eigenfrequencies and a uniquely identifiable solution, thereby avoiding mode-jump behavior.

4.2.2 Task 2: Strike Position Recovery

Signal-level template matching achieves sub-millimeter accuracy ($0.3\% \times R \approx 0.4$ mm) across all shapes and radial positions (Table 3), with zero error at the center reflecting the strong, distinct modal signature of central strikes. A slight degradation appears near the boundary for the oval case ($0.9\%R$ at $r = 0.90R$), likely due to reduced modal sensitivity near the rim, but the overall mean error is unaffected ($0.3\%R$). These results confirm that signal-domain template matching provides a robust, geometry-invariant localization mechanism well-suited to varying structural configurations.

4.2.3 Task 3: Shape Identification

Shape identification via modal basis projection achieves highly accurate classification across all test cases (Table 4): the R^2 confusion matrix is strongly diagonally dominant (diagonal values of 0.90, off-diagonal values $\leq$ 0.13). The modestly elevated cylinder–hourglass cross-response (0.130) reflects their partial geometric similarity, while the oval's near-zero cross-correlation with other shapes reflects its lifted modal degeneracies. Modal basis projection thus provides a robust, discriminative representation for shape identification.

Table 4: Shape identification: R^2 values for three shapes. Diagonal dominance indicates perfect classification.

True Shape	Cylinder	Hourglass	Oval	Identified
Cylinder	0.900	0.005	0.055	Cylinder ✓
Hourglass	0.130	0.900	0.035	Hourglass ✓
Oval	0.004	0.001	0.900	Oval ✓

4.3 Validation on Real Chenda Recordings

4.3.1 Dataset and Preprocessing

We evaluate PGSD on real recordings of the Chenda drum. The dataset comprises strikes at three anatomical locations (center, edge, hand) with three intensity levels (low, medium, high), spanning multiple recording sessions. In total, the dataset includes 50 recordings, which are segmented into 50 individual strike events using onset detection [22].

To isolate the modal decay phase, the initial impact transient is removed by detecting the onset peak and extracting a 3.0 s window following each strike. The signals are then mean-centered and amplitude-normalized. To suppress non-modal high-frequency components introduced by stick impact and recording noise, a low-pass filter with cutoff frequency ≈ 2 kHz is applied, chosen to lie above the frequency range where the retained modal basis contributes meaningful energy while discarding impact transients and stick noise. This preprocessing step is important for improving model-data alignment when applying physics-derived modal bases. The dataset spans multiple strike types and intensities, providing a representative evaluation of real-world variability under controlled recording conditions.

4.3.2 Performance Comparison

Applying physics-derived modal bases directly to real recordings yields moderate reconstruction accuracy, since modelled and observed modal frequencies differ due to unknown physical parameters (e.g., tension, boundary conditions) and performer-dependent excitation. Initializing modal frequencies from each signal's dominant spectral peaks improves reconstruction quality. Performance is evaluated by comparing the original signal $\mathbf{x} \in \mathbb{R}^K$ with its model-based reconstruction $\hat{\mathbf{x}}$, resynthesized from each method's estimated parameters using its corresponding signal model (sinusoidal components, spectral bases, or modal functions).

We quantify reconstruction accuracy using the coefficient of determination (R^2), defined as:

$$R^2 = 1 - \frac{\|\mathbf{x} - \hat{\mathbf{x}}\|_2^2}{\|\mathbf{x}\|_2^2}, \tag{13}$$

which measures the proportion of signal energy explained by the model. An R^2 value of 1 indicates perfect reconstruction, while lower values reflect increasing mismatch between the original and reconstructed signals [23, 24].

We compare PGSD against classical spectral and signal modeling baselines [15, 17]: FFT-based peak tracking, which captures dominant spectral components but lacks temporal and modal structure; sinusoidal modeling, which

Table 5: Comparison of reconstruction performance and perceptual quality across methods.

Method	Type	Mean R^2	MOS
FFT Peak Tracking	Spectral baseline	0.45	2.49 ± 0.07
Sinusoidal Modeling	Signal model (global)	0.59	3.08 ± 0.17
NMF	Data-driven	0.41	1.86 ± 0.15
PGSD (proposed)	Physics-guided	**0.70**	$\mathbf{3.69 \pm 0.10}$

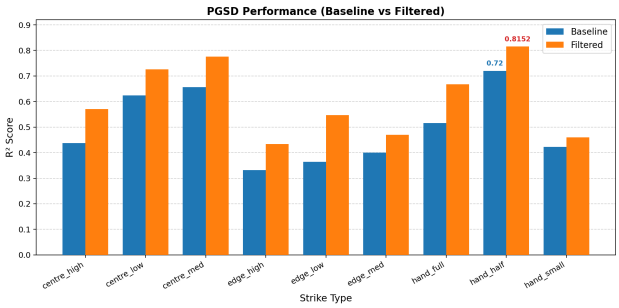


Figure 4: PGSD across nine strike types before and after spectral filtering. It improves performance, with larger gains observed for edge-based strikes.

omits mode-dependent damping; and NMF [25], which achieves moderate accuracy ($R^2 = 0.41$) without yielding physically interpretable parameters. As shown in Table 5, PGSD achieves the highest reconstruction accuracy among all methods (mean $R^2 \approx 0.70$) while additionally enabling interpretable parameter estimation and controllable synthesis.

Fig. 4 summarizes reconstruction accuracy (R^2) across strike types before and after spectral filtering, which improves accuracy from 0.65 to 0.70 on average (peak $R^2 = 0.8152$ for hand-half strikes). Gains are largest for high-frequency-excited strikes (e.g., *edge_low*, $+0.13$ R^2) and smallest for smoother hand-based strikes ($+0.01$–0.09 R^2), consistent with the modal model's assumption of low-frequency, smoothly decaying excitation.

4.3.3 Perceptual Evaluation

To complement R^2 metrics, a MOS test was conducted with 20 listeners rating reconstructions on a 1–5 scale for 50 audio files. As reported in Table 5, PGSD achieves the highest perceptual quality among reconstruction methods (MOS = 3.69), followed by Sinusoidal modeling (3.08), FFT (2.49), and NMF (1.86), consistent with the objective R^2 ranking in Table 5.

4.3.4 Qualitative Validation of Strike Position

We perform qualitative validation using labelled real-recording strike categories (center, hand, edge). As shown in Table 6, estimated radii closely match expected locations (absolute error < 0.06 in r/R), with correct ordinal separation (center 0.22 ± 0.05 vs. 0.20; hand 0.45 ± 0.05 vs. 0.46; edge 0.81 ± 0.05 vs. 0.75), and inter-class differences well above the intra-class variability (≈ 0.05). A mild positive bias for edge strikes is consistent with reduced modal sensitivity near the rim. These results provide indirect validation that estimated positions align with known excitation patterns.

Table 6: Estimated normalized strike radius across categories ($R = 1$ corresponds to the membrane edge).

Category	Estimated (r/R)	Ground Truth (r/R)
Center	0.22 ± 0.05	0.20
Hand	0.45 ± 0.05	0.46
Edge	0.81 ± 0.05	0.75

4.4 Discussion

By constructing sound directly from modal dynamics, the proposed framework provides a digital twin representation in which acoustic behavior can be explored through parameter manipulation. The bidirectional design unifies synthesis and analysis within a shared representation: the forward model enables controlled simulation, while PGSD aligns the model with real recordings. This framework has broader implications beyond MIR, including virtual instrument design with interpretable controls, scientific analysis of percussion acoustics, and data-efficient modeling in low-resource settings. Although performance on real recordings is affected by model–data mismatch, consistent behavior across strike types suggests that the model captures meaningful physical structure.

The present evaluation is limited to classical spectral and signal-based baselines (FFT peak tracking, sinusoidal modeling, NMF); comparison against transfer-function-based modal synthesis [8] and differentiable or neural synthesis approaches [10, 11] remains an important direction for future work.

Reproducibility. Implementation, code, and dataset annotations, along with audio examples of synthesis and real-recording reconstruction, are available at: `https://github.com/arcadehero3-eng/CHENDA_PGSD.git`.

5. CONCLUSION

We presented a physics-based parametric synthesis framework for percussive instruments, enabling controllable sound generation through direct manipulation of physical parameters. By modeling membrane dynamics via eigenmodes, the system acts as a digital twin linking structure and sound. PGSD provides the inverse counterpart for calibration and parameter inference, forming a unified bidirectional framework. Results demonstrate physically consistent behavior and good alignment with real recordings despite model mismatch. Modal degeneracy is identified as a key limitation, mitigated through asymmetric loading. This work highlights physics-based representations as an interpretable alternative to data-driven approaches. Future work includes improved excitation modeling, hybrid physics–learning methods, and extension to broader instrument classes.

6. AI USAGE STATEMENT

AI-based tools were used to assist in debugging and correcting code implementing the physics-based model (Section 2). All methodology, experimental design, analysis, and conclusions are the authors' own; AI-assisted code was reviewed and verified by the authors before use in the reported experiments. AI tools were also used for language editing of the manuscript text.

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